

Re=1000 Audit: the base regime on sequences no model ever trained on

Baseline inference, the dials alone and coupled, the depth ladder, and the fine-tuned variants. Fixed K3x86 unless stated. Held-out sequences 34-39 of the 40-seed file (base pretrain 0-31, val 32-35 with the fine-tunes on 32-33, test 36-39); 20 frames each. 2026-08-25.

Provenance note

The earlier cross-regime grid evaluated Re=1000 on sequences 4-19, which are base-TRAINING sequences; this audit supersedes those numbers. The measured memorization inflation was small (base K3 unguided: 0.48/0.92 there vs 0.46/0.91 here), so no earlier conclusion flips, but every Re=1000-specific number in the write-up should come from this audit.

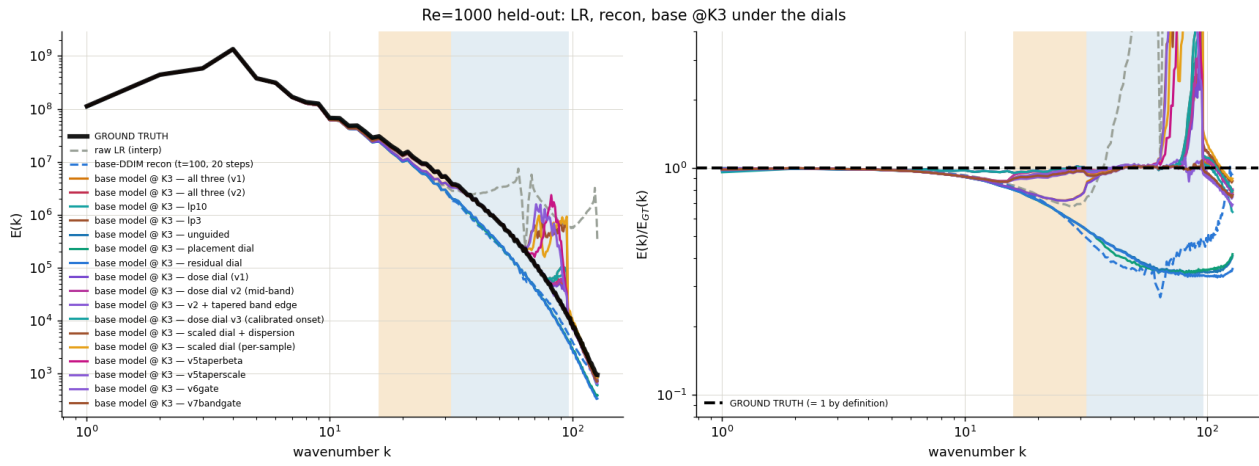
1 · The observation and the recon

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place /triplet	place pooled	MSE	resid
raw LR	0.99	0.95	0.76	2.37	20.62	0.635	0.669	1.28	370.58
base-DDIM recon	0.99	0.95	0.71	0.41	0.38	0.755	0.745	0.37	0.82

The LR's above-Nyquist "retention" (2.4x / 21x) is aliased excess, not content. The recon strips it, keeps 0.41 of the fine band, places at 0.745, and has the best MSE of any row - the well-placed conditional mean.

2 · The base model under the dials, alone and coupled

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place /triplet	place pooled	MSE	resid
base K3 all three v1	0.99	0.94	0.76	0.95	1.16	0.859	0.741	0.43	1.51
base K3 all three v2	0.99	0.95	0.97	0.99	1.18	0.864	0.732	0.46	1.62
base K3 lp10	0.99	0.98	0.97	0.99	1.24	0.905	0.750	0.49	1.34
base K3 lp3	0.99	0.98	0.97	0.99	1.38	0.908	0.740	0.49	2.13
base K3 unguided	0.99	0.94	0.71	0.46	0.35	0.908	0.908	0.40	0.62
base K3 placement	0.99	0.94	0.70	0.45	0.35	0.819	0.855	0.41	0.65
base K3 residual	0.99	0.94	0.71	0.46	0.34	0.907	0.907	0.40	0.49
base K3 dose v1	0.99	0.94	0.76	0.95	1.13	0.910	0.753	0.43	1.62
base K3 dose v2	0.99	0.95	0.97	0.99	1.18	0.912	0.750	0.46	1.82
base K3 tapered	0.99	0.98	0.97	1.00	1.03	0.911	0.749	0.49	0.83
base K3 v3 calibrated	0.99	0.98	0.97	0.99	1.38	0.908	0.740	0.49	2.13
base K3 v4beta	0.99	0.95	0.94	0.98	6.79	0.898	0.468	0.48	18.98
base K3 v4scaled	0.99	0.95	0.94	0.98	7.00	0.894	0.446	0.49	20.26
base K3 v5taperbeta	0.99	0.95	0.94	0.99	7.91	0.903	0.304	0.50	16.40
base K3 v5taperscale	0.99	0.95	0.94	0.99	8.00	0.904	0.359	0.50	14.43
base K3 v6gate	0.99	0.95	0.91	0.96	1.01	0.907	0.908	0.41	1.00
base K3 v7bandgate	0.99	0.95	0.93	0.96	1.01	0.907	0.907	0.41	1.01

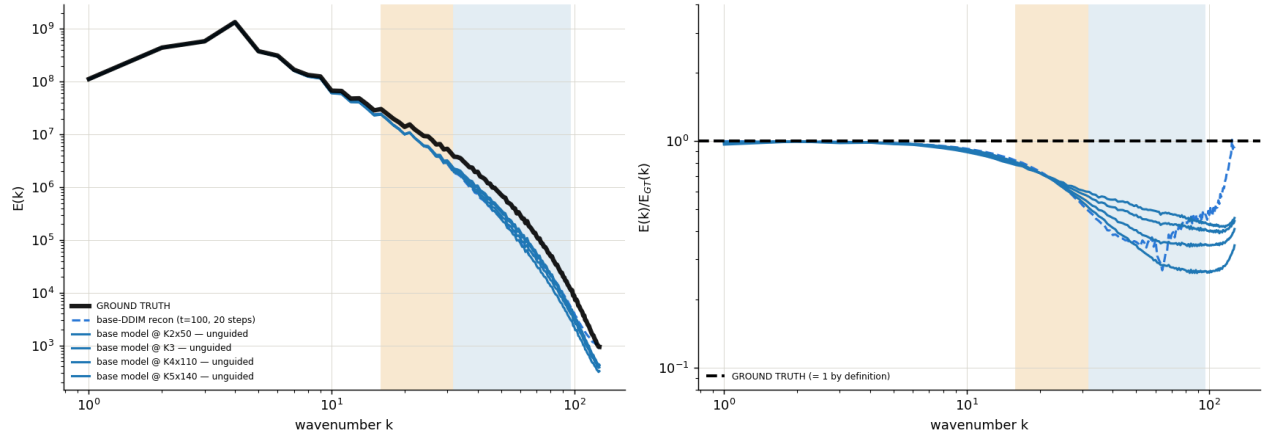


In-regime every dose-adding dial takes the base to ~ 1.0 retention but pushes its residual above the true flow's level (1.6-1.9x) and costs 0.15 placement; the residual dial polishes; the placement dial HURTS the base here (0.91 to 0.86 - the LR-map ceiling); v2 closes the mid band (0.71 to 0.96).

3 · Inference depth: the base's unsteered ladder

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place /triplet	place pooled	MSE	resid
base K2x50 unguided	0.99	0.95	0.71	0.42	0.28	0.905	0.900	0.37	0.57
base K3 unguided	0.99	0.94	0.71	0.46	0.35	0.908	0.908	0.40	0.62
base K4x110 unguided	0.99	0.94	0.71	0.51	0.43	0.892	0.901	0.48	0.66
base K5x140 unguided	0.98	0.93	0.71	0.55	0.47	0.874	0.883	0.59	0.67

Re=1000 held-out: the base model's unsteered depth ladder

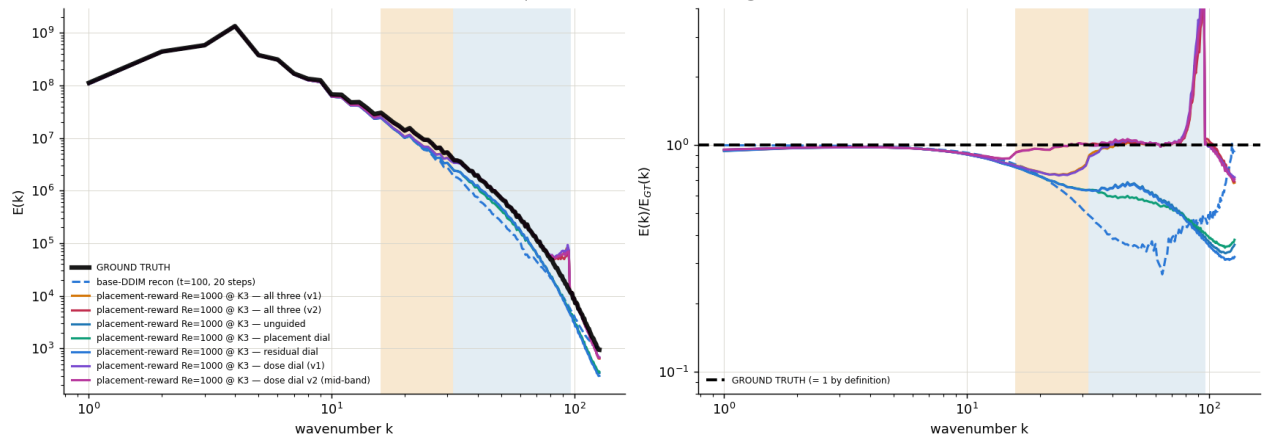


K2 to K5: retention 0.42 to 0.55 only, placement slipping, MSE rising - every rung sags to 0.25-0.45 through the fine band. Depth is the wrong lever in-regime.

4 · placement-reward Re=1000 (pr1k-549)

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place /triplet	place pooled	MSE	resid
placement-reward Re=1000 K3 all three v1	0.98	0.94	0.77	0.97	1.23	0.834	0.729	0.48	1.87
placement-reward Re=1000 K3 all three v2	0.98	0.95	0.96	1.01	1.20	0.840	0.714	0.52	1.90
placement-reward Re=1000 K3 unguided	0.98	0.94	0.72	0.64	0.52	0.796	0.779	0.46	0.98
placement-reward Re=1000 K3 placement	0.98	0.94	0.72	0.60	0.51	0.817	0.840	0.45	1.03
placement-reward Re=1000 K3 residual	0.97	0.94	0.72	0.64	0.51	0.796	0.778	0.46	0.79
placement-reward Re=1000 K3 dose v1	0.98	0.94	0.77	0.98	1.28	0.796	0.676	0.49	2.12
placement-reward Re=1000 K3 dose v2	0.98	0.95	0.96	1.02	1.24	0.813	0.681	0.52	2.04

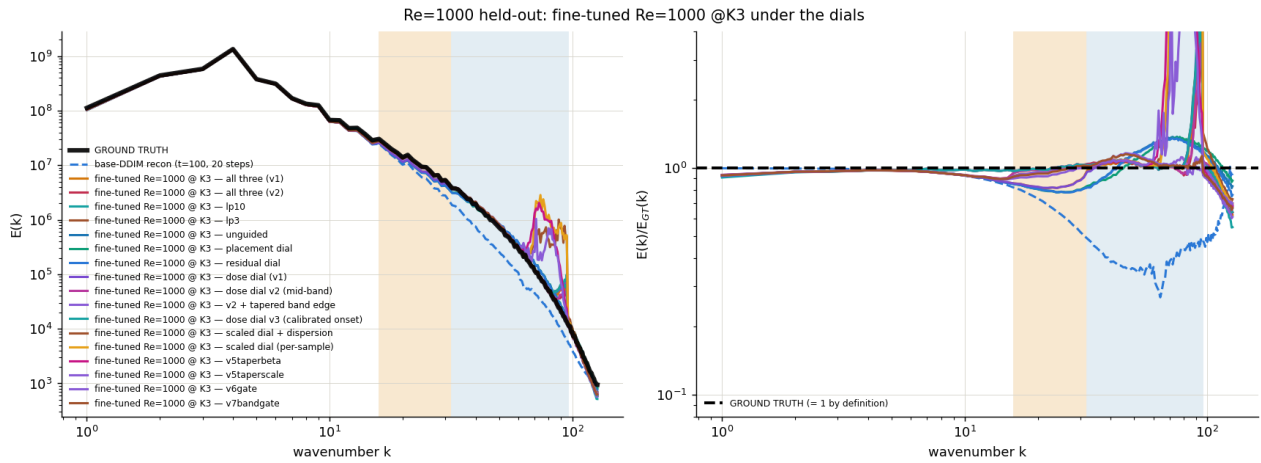
Re=1000 held-out: placement-reward Re=1000 @K3 under the dials



5 · ft Re=1000 (r1k-449)

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place /triplet	place pooled	MSE	resid
ft Re=1000 K3 all three v1	0.97	0.95	0.84	1.02	1.15	0.844	0.739	0.45	1.33

ft Re=1000 K3 all three v2	0.97	0.95	0.99	1.05	1.08	0.853	0.726	0.48	1.20
ft Re=1000 K3 lp10	0.97	0.98	0.99	1.05	1.23	0.854	0.730	0.51	1.33
ft Re=1000 K3 lp3	0.97	0.98	0.99	1.05	1.19	0.859	0.736	0.51	1.31
ft Re=1000 K3 unguided	0.97	0.95	0.81	0.97	1.33	0.856	0.863	0.43	0.79
ft Re=1000 K3 placement	0.97	0.95	0.81	0.94	1.32	0.842	0.863	0.43	0.86
ft Re=1000 K3 residual	0.97	0.95	0.81	0.96	1.31	0.853	0.862	0.43	0.64
ft Re=1000 K3 dose v1	0.97	0.95	0.84	1.03	1.18	0.853	0.740	0.45	1.47
ft Re=1000 K3 dose v2	0.97	0.95	0.99	1.06	1.09	0.862	0.736	0.48	1.32
ft Re=1000 K3 tapered	0.97	0.98	0.99	1.05	1.04	0.859	0.732	0.51	0.68
ft Re=1000 K3 v3 calibrated	0.97	0.98	0.99	1.05	1.19	0.859	0.736	0.51	1.31
ft Re=1000 K3 v4beta	0.97	0.95	0.96	1.05	6.86	0.858	0.513	0.51	18.56
ft Re=1000 K3 v4scaled	0.97	0.95	0.96	1.05	13.24	0.856	0.211	0.58	31.80
ft Re=1000 K3 v5taperbeta	0.97	0.95	0.96	1.05	9.68	0.858	0.267	0.54	17.13
ft Re=1000 K3 v5taperscale	0.97	0.95	0.96	1.06	3.90	0.860	0.682	0.47	6.32
ft Re=1000 K3 v6gate	0.97	0.95	0.93	1.08	1.03	0.863	0.874	0.43	0.78
ft Re=1000 K3 v7bandgate	0.97	0.95	0.95	1.08	1.03	0.866	0.874	0.43	0.80



Fine-tune vs steering, in-regime: the r1k fine-tune reaches the same retention as base+dial (0.99 vs 1.00) while placing the energy (+0.11) and keeping the flow physical (residual 0.79 below the true level vs 1.82 above it). Putting the dial on top of a well-trained in-regime model is harmful (placement -0.13, residual x1.7). Doctrine: fine-tune at home, steer away from home.

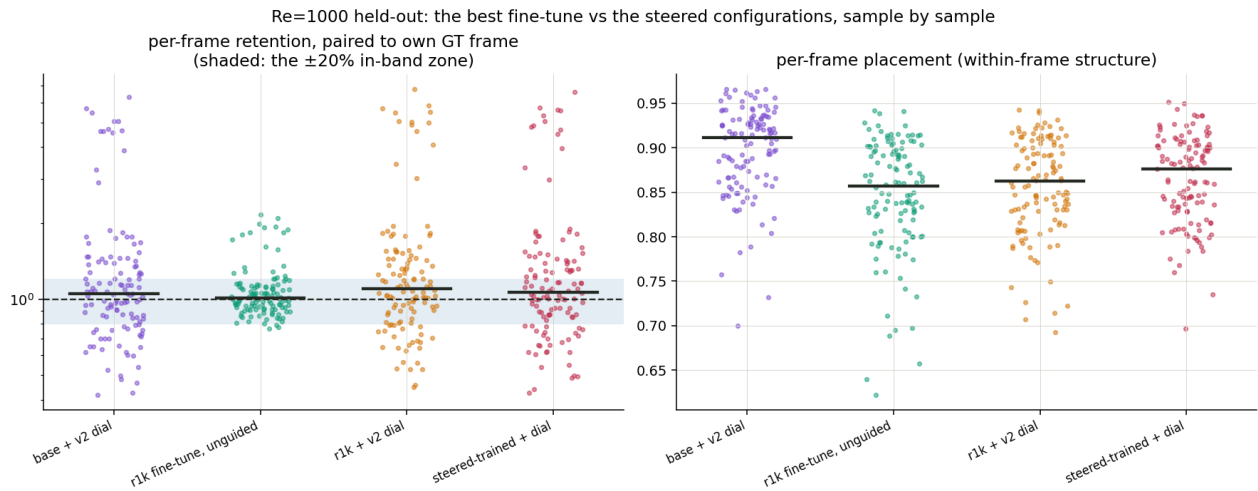
Dial-refinement ablation (each fix in isolation on top of v2)

config	ret[16,32]	ret[32,64]	ret[64,96]	place	pl[64,96]	resid	MSE
base: v2 baseline	0.97	0.99	1.18	0.750	0.80	1.82	0.46
base: + tapered edge	0.97	1.00	1.03	0.749	0.80	0.83	0.49
base: + onset (v3)	0.97	0.99	1.38	0.740	0.80	2.13	0.49
base: + physics x10	0.97	0.99	1.24	0.750	0.79	1.34	0.49
base: + per-sample scale	0.94	0.98	7.00	0.446	-	20.26	0.49
base: + scale, dispersion	0.94	0.98	6.79	0.468	-	18.98	0.48
config	ret[16,32]	ret[32,64]	ret[64,96]	place	pl[64,96]	resid	MSE
r1k-449: v2 baseline	0.99	1.06	1.09	0.736	0.72	1.32	0.48
r1k-449: + tapered edge	0.99	1.05	1.04	0.732	0.72	0.68	0.51
r1k-449: + onset (v3)	0.99	1.05	1.19	0.736	0.72	1.31	0.51
r1k-449: + physics x10	0.99	1.05	1.23	0.730	0.72	1.33	0.51
r1k-449: + per-sample scale	0.96	1.05	13.24	0.211	-	31.80	0.58
r1k-449: + scale, dispersion	0.96	1.05	6.86	0.513	-	18.56	0.51

The taper is decisive in-regime: it removes the $k \sim 95$ pile-up at source (deep band 1.38 to 1.03, residual 2.13x to 0.83x - physical again, deep placement 0.57 to 0.62). Physics weight $\times 10$ treats the symptom and costs placement (0.41); $\times 30$ diverges. The onset term is counterproductive in-regime on the base. Per-sample scaling targets the orthogonal frame-amplitude collapse.

The head-to-head: best fine-tune vs the steered configurations, per sample

	ens ret	ens place	resid	ps ret med [p10,p90]	in-band	ps place
base + v2 dial	1.00	0.750	1.82	1.05 [0.65,2.93]	38%	0.912
r1k unguided	0.99	0.863	0.79	1.01 [0.85,1.38]	78%	0.856
r1k + v2 dial	1.06	0.736	1.32	1.10 [0.67,3.05]	38%	0.862
st1k + dial	1.03	0.729	2.29	1.07 [0.65,3.01]	38%	0.876



Only the plain fine-tune validates per sample (78% of frames within $\pm 20\%$ of their own frame's energy vs 38% for every dialed configuration): the fine-tune learned *CONDITIONAL* dose, the dial imposes the *MARGINAL* one. And the pooled placement metric conflates within-frame structure (steered base best, 0.912) with frame-to-frame amplitude consistency (dial worst) - the fine-tune's pooled advantage is mostly amplitude discipline. In-regime, r1k unguided is the only configuration well-dosed per frame, structurally sound, and physical.

Per-sample validation

row	ret paired med [p10,p90]	place/frame med [p10,p90]	MSE med	frames in [0.8,1.2]
raw LR	3.39 [2.64,7.23]	0.635 [0.517,0.746]	1.24	0%
base K2x50 unguided	0.42 [0.37,0.48]	0.905 [0.830,0.943]	0.36	0%
base K3 all three v1	0.99 [0.62,2.73]	0.859 [0.780,0.905]	0.41	41%
base K3 all three v2	1.05 [0.63,2.93]	0.864 [0.793,0.912]	0.44	39%
base K3 lp10	1.06 [0.64,2.91]	0.905 [0.834,0.949]	0.45	38%
base K3 lp3	1.04 [0.63,2.93]	0.908 [0.838,0.947]	0.45	39%
base K3 unguided	0.46 [0.41,0.54]	0.908 [0.840,0.942]	0.40	0%
base K3 placement	0.44 [0.39,0.52]	0.819 [0.704,0.888]	0.40	0%
base K3 residual	0.45 [0.41,0.54]	0.907 [0.840,0.941]	0.40	0%
base K3 dose v1	1.00 [0.63,2.74]	0.910 [0.827,0.947]	0.41	41%
base K3 dose v2	1.05 [0.65,2.93]	0.912 [0.839,0.948]	0.43	38%
base K3 tapered	1.05 [0.64,2.97]	0.911 [0.839,0.948]	0.44	38%
base K3 v3 calibrated	1.04 [0.63,2.93]	0.908 [0.838,0.947]	0.45	39%
base K3 v4beta	1.03 [0.90,2.24]	0.898 [0.711,0.948]	0.43	78%
base K3 v4scaled	1.03 [0.90,3.50]	0.894 [0.664,0.948]	0.43	78%
base K3 v5taperbeta	1.01 [0.87,1.41]	0.903 [0.790,0.952]	0.41	88%
base K3 v5taperscale	1.01 [0.88,1.40]	0.904 [0.813,0.953]	0.42	86%
base K3 v6gate	0.97 [0.85,1.12]	0.907 [0.839,0.954]	0.39	94%
base K3 v7bandgate	0.98 [0.86,1.12]	0.907 [0.834,0.954]	0.40	94%
base K4x110 unguided	0.50 [0.45,0.61]	0.892 [0.813,0.940]	0.48	0%
base K5x140 unguided	0.54 [0.47,0.66]	0.874 [0.765,0.932]	0.60	0%
placement-reward Re=1000 K3 all three v1	1.03 [0.64,2.77]	0.834 [0.740,0.896]	0.46	41%
placement-reward Re=1000 K3 all three v2	1.06 [0.64,3.03]	0.840 [0.761,0.898]	0.49	38%
placement-reward Re=1000 K3 unguided	0.63 [0.53,0.75]	0.796 [0.637,0.907]	0.45	6%
placement-reward Re=1000 K3 placement	0.59 [0.49,0.70]	0.817 [0.708,0.878]	0.44	2%
placement-reward Re=1000 K3 residual	0.63 [0.53,0.75]	0.796 [0.637,0.907]	0.45	5%
placement-reward Re=1000 K3 dose v1	1.03 [0.65,2.77]	0.796 [0.642,0.909]	0.47	40%
placement-reward Re=1000 K3 dose v2	1.08 [0.65,3.00]	0.813 [0.669,0.915]	0.50	38%
ft Re=1000 K3 all three v1	1.07 [0.68,2.87]	0.844 [0.746,0.898]	0.43	40%
ft Re=1000 K3 all three v2	1.10 [0.67,3.09]	0.853 [0.771,0.901]	0.46	38%
ft Re=1000 K3 lp10	1.11 [0.67,3.06]	0.854 [0.781,0.918]	0.47	37%
ft Re=1000 K3 lp3	1.11 [0.68,3.08]	0.859 [0.790,0.920]	0.47	39%
ft Re=1000 K3 unguided	1.01 [0.85,1.38]	0.856 [0.773,0.914]	0.42	78%
ft Re=1000 K3 placement	0.98 [0.84,1.38]	0.842 [0.735,0.895]	0.42	81%
ft Re=1000 K3 residual	1.00 [0.85,1.38]	0.853 [0.771,0.913]	0.42	78%
ft Re=1000 K3 dose v1	1.08 [0.69,2.90]	0.853 [0.775,0.918]	0.43	38%
ft Re=1000 K3 dose v2	1.10 [0.67,3.05]	0.862 [0.789,0.917]	0.45	38%
ft Re=1000 K3 tapered	1.10 [0.67,3.09]	0.859 [0.786,0.920]	0.47	37%
ft Re=1000 K3 v3 calibrated	1.11 [0.68,3.08]	0.859 [0.790,0.920]	0.47	39%
ft Re=1000 K3 v4beta	1.08 [0.93,3.06]	0.858 [0.472,0.919]	0.44	75%
ft Re=1000 K3 v4scaled	1.08 [0.93,2.04]	0.856 [0.584,0.921]	0.44	75%

ft Re=1000 K3 v5taperbeta	1.07 [0.91,1.46]	0.858 [0.678,0.920]	0.44	80%
ft Re=1000 K3 v5taperscale	1.07 [0.92,1.36]	0.860 [0.742,0.921]	0.44	78%
ft Re=1000 K3 v6gate	1.07 [0.95,1.26]	0.863 [0.763,0.921]	0.42	83%
ft Re=1000 K3 v7bandgate	1.08 [0.95,1.25]	0.866 [0.779,0.921]	0.42	82%
base-DDIM recon	0.41 [0.36,0.48]	0.755 [0.585,0.865]	0.35	0%

